# Engine Architecture

## Engine Architecture I

The engine is nine small modules, each with one responsibility.

`geometry` is pure arithmetic over points: the page trim, its margins, and a `ColumnGrid` that partitions a region into equal columns with gutters and reports the centre of each gutter (where a hairline rule is drawn). It imports no rendering library, which makes the column mathematics trivially testable.

`content` defines the typed object graph — `Edition`, `Page`, `Story`, `Box`, `Figure`, `Block` — and the strict YAML loaders that build it. The loaders are forgiving about *shape* (a body element may be a bare string or a tagged mapping) but strict about *structure*: an unknown page template or box kind raises an error naming the offending value and the allowed set.

`config` is the strict render configuration (trim size, default column count, rail width, gutter), validated the same way.

`typography` owns every font name: it registers the Didot display, Georgia text and Helvetica label faces (with a base-14 fall-back when the system lacks them) and builds the paragraph stylesheet that the rest of the engine draws from through named `Fonts` roles.

`figures` is the only module that *generates* imagery: the halftone harbour, lighthouse and redwood engravings (Pillow), the weather, tide and crab-landings charts (Matplotlib), and the colour display-ad graphics. It writes deterministic PNGs that `components` later places.

`components` builds the flowables that live *inside* a column: a story's headline, byline and body; a raised drop cap; a pull quote; a ruled box; a data table; a figure with its cutline. `furniture` draws the elements that are *placed* rather than flowed: the nameplate and its ears, the section banner, the folio, the hairline column rules, and the spanning lead headline.

`layout` is the bridge. For each page it draws the furniture (which returns the y-coordinate where the column grid begins), constructs the column frames — including an optional narrow rail and the room reserved for a spanning headline — and pours whole flowables into the frames with ReportLab's `Frame.addFromList` without paragraph splitting. Heading-and-opening groups move together to the next column. Content that does not fit is returned and reported as an *over-set* page rather than silently dropped.

`engine` ties it together: register fonts, build the stylesheet, open a canvas at the configured trim, render each page, and write the PDF plus a machine-readable render report.

This **drawn-furniture / flowed-body** split is the key architectural choice. A pure document-template approach cannot place a headline that spans several columns above body text that flows beneath it; drawing the furniture first, and beginning the column frames below it, makes spanning headlines, standing boxes and automatic text flow coexist on the same page.

# Methods I

Producing an edition is a deterministic, four-step method, run by the pipeline and covered end-to-end by the test suite:

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1. **Load** — `content` parses `edition.yaml` and every `content/pages/*.yaml` into the typed `Edition` graph, rejecting unknown page templates or box kinds by naming the offending value and the allowed set.
2. **Measure** — `geometry` derives the trim, margins and `ColumnGrid` for each page from the strict `config`, independently of any rendering library.
3. **Compose** — `components` and `furniture` build the flowed and drawn elements; `layout` pours them into the column frames and reports any content that does not fit as an over-set page rather than dropping it silently.
4. **Emit** — `engine` registers fonts, renders each page to the canvas, and writes the PDF alongside a machine-readable render report.

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Every step is pure-data-in, artifact-out and seeded deterministically, so the same edition manifest always yields a byte-identical paper — the reproducibility contract verified in `tests/` and described in `04_reproducibility.md`.